

A Non-Invasive Acoustic Sensing System for Bruxism Screening

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Abstract—Sleep bruxism (teeth grinding) is a common, frequently undiagnosed oral disorder, and is currently screened using expensive sleep labs or uncomfortable EMG-based devices. This study explores whether a low-cost, in-ear MEMS microphone can detect teeth grinding from acoustic signals and distinguish it from ambient noise. Occlusion-effect testing found that in-ear microphone placement increased grinding-signal amplitude by roughly 10× relative to a near-jaw placement. Using 30-minute grinding recordings under silence and fan noise, a 20–80 Hz frequency band was identified as unique with grinding after excluding peaks shared with non-grinding baseline recordings. A MATLAB-based filtering algorithm using envelope thresholding, gap-bridging, duty-cycle, and spectral-flatness criteria correctly classified all tested trials (3/3 No Warning, 3/3 Mild, 2/2 Severe) for a sample size of one (N=1). Results support the feasibility of audio-only bruxism screening, though generalization beyond one subject and validation against polysomnography metrics remain future work.

I. INTRODUCTION

A. Background Information

Bruxism (teeth grinding) is a frequently undiagnosed oral disorder that can lead to serious dental damage, poor sleep, and long-term health issues.

- Bruxism (awake or sleep) affects an estimated 24% of adults and 19% of children.
- Sleep bruxism specifically affects an estimated 21% of the global population.

Common causes of bruxism include stress and anxiety, abnormal tooth alignment, certain medications, neurological conditions, and substance abuse. Symptoms include tooth wear and sensitivity, temporomandibular joint (TMJ) pain and stiffness, recurrent headaches or migraines, discomfort in biting or chewing, and poor sleep quality. If untreated over months or years, irreversible damage may occur.

Current diagnosis requires expensive overnight sleep lab stays (polysomnography, or PSG), and existing wearable devices that are uncomfortable and impractical for daily use primarily because movement during sleep disrupts electrode placement in EMG-based wearables.

B. Research Problem or Question

The overarching question is: can bruxism be accurately detected using acoustic signals captured from a patient as they

sleep overnight, and can this be accomplished through a low-cost wearable?

This research investigates whether a small microphone placed inside the ear canal can reliably capture the acoustic signatures of oral events and whether teeth grinding frequencies can be distinguished from ambient environmental noise. Following that, the research explores the possibility for an algorithm that flags the severity of teeth grinding based on the audio recordings. It will also lay the groundwork for a low-cost, non-invasive, at-home earable device that could screen for bruxism without requiring clinical settings.

C. Significance of the Research

An audio only approach to bruxism screening has clear practical advantages over existing options, as it avoids the electrode-placement problems tied to EMG sensors, and makes at-home screening accessible at a scale that PSG and EMG-based devices cannot match because it is built with low cost, off-the-shelf components. If validated further, this approach could allow for remote monitoring to help track the effects of medication over time, and support early, preventative detection before dental damage occurs. As shown in the device comparison in Section II, most current commercial screening devices remain EMG-based, and this study's fully audio-only approach remains distinctive within that landscape, addressing a gap that has been largely unaddressed to date.

II. LITERATURE REVIEW

A. Overview of Relevant Literature

The current clinical gold standard for diagnosing sleep bruxism is polysomnography (PSG), typically performed with audio-visual recording (PSG-AV) in a sleep lab. While accurate, PSG is expensive, inconvenient for patients, and not scalable for widespread or repeated screening.

Several ambulatory devices have been developed as PSG alternatives. All are EMG based, detecting bruxism via electromyographic, or muscle-activity, signals. Two of the seven devices reviewed in Table I (NOX T3 and Sleep Profiler) additionally include audio recording, but none rely on audio as their primary bruxism detection signal in the way this study's earable design does.

TABLE I
Comparison of Current Commercially Available Ambulatory Bruxism-Screening Devices

Category	Bitestrip	dia-BRUXO	GrindCare	Bruxoff	NOX T3	Sleep Profiler	Eearable (this study)
EMG-based	X	X	X	X	X	X	—
Audio recording	—	—	—	—	X	X	X
Multi-night capable	—	—	X	X	—	—	X
Sensitivity / Specificity vs. PSG-AV	72–84.2% / —	Not evaluated	60% / 60%	83.3% / 72.2%	48.3% / 81.2%	100% / 93.3%	Not evaluated (this study)

Source: Australian Dental Journal (2024)

Polysomnography remains the reference standard but shares the same limitation across all alternatives: expense, inconvenience, and poor scalability for daily use.

B. Key Theories or Concepts

The occlusion effect is an acoustic phenomenon in which sealing the ear canal increases the perceived loudness of bone-conducted, low-frequency sound relative to an open ear. This is because sealing the ear traps and reinforces vibration conducted through the skull and jaw rather than sound traveling through open air (Acta Acustica, 2021).

Related literature supports the physiological basis for this study's acoustic sensing approach. Jaw movement produces bone-conducted acoustic sound that can be captured near the ear canal. Separately, snoring has been shown to produce acoustic signatures that are distinguishable from bruxism grinding, and at least one existing patent describes a device made specifically to sense teeth grinding acoustic signals and to differentiate them from other noise (PubMed, 2025; U.S. Patent, 2025).

Together, this predicts that an in-ear (occluded) microphone should better capture jaw-generated, bone-conducted sounds such as grinding and snapping. This study's own occlusion-effect trial (Sections III and IV) was designed to test this prediction directly for grinding.

III. METHODOLOGY

A. Research Design

An Arduino-interfaced MEMS microphone was used to record acoustic signals near and within the ear canal (Fig. 0). Data collection proceeded in two phases.

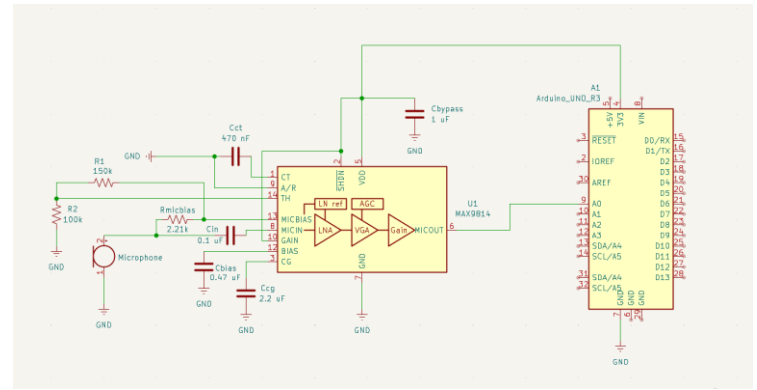


Fig. 0. Arduino and MEMS microphone circuit used for all recordings (MAX9814 microphone amplifier with automatic gain control, feeding an Arduino Uno analog input).

Phase 1 — Microphone Placement / Occlusion Effect Trial

To determine the optimal microphone placement, three trials of 10 seconds of teeth grinding each were recorded under two placement conditions:

- the microphone positioned close to the jaw, outside the ear
- the microphone inserted within the ear canal (occluded)

The in-ear placement captured a stronger grinding signal than the near-jaw placement, showing a great increase of roughly 10× in signal amplitude, which is consistent with the occlusion effect described in Section II. Based on this result, the in-ear (occluded) placement was adopted for all subsequent data collection, and additionally offered a more comfortable, secure fit for prolonged wear.

Phase 2 — Grinding and Ambient Noise Characterization

With microphone placement fixed to in-ear, 30 minutes of teeth grinding was recorded under each of two conditions: silence and fan noise. To separate grinding-related frequency content from unrelated background content, an additional 5 minutes of silence only and 5 minutes of fan-noise only baseline audio, with the microphone in the ear, was also collected. To get the desired frequency range, a sample frequency of 5000 Hz was used. Frequency peaks that appeared in both the grinding recordings and the no grinding baseline were treated as artifacts unrelated to grinding and were deemed to likely be powerline/electrical interference and its harmonics.

B. Data Collection & Sample Selection

The researcher/developer served as the only subject ($N = 1$). Therefore, this study is best understood as an exploratory feasibility study rather than a clinically validated trial. Its purpose is to establish whether acoustic grinding signatures are detectable and distinguishable under controlled conditions, providing a foundation for later multiple subject validation rather than an immediate generalizable model (see Limitations, Section V).

C. Data Analysis Techniques

Data was collected through the Arduino and stored in Excel, then imported into MATLAB. A MATLAB program applied a Fourier Transform and, for display, Welch's method for power spectral density estimation, to convert the acoustic signal into the frequency domain. This was done to identify the dominant frequency components associated with grinding and snapping, and breathing events, and to compare grinding recordings against the silence/fan baselines described above.

Filtering Algorithm Design

Building on the identified grinding frequency band of 20-80 Hz (Section IV), a MATLAB-based classification algorithm was

developed to categorize each recording into one of three warning levels: No Warning, Mild Warning, or Severe Warning. The algorithm was evaluated using recordings with known, self-labeled ground truth, consisting of three trials each for each type of warning. The No Warning recordings consisted of normal speech, humming, and little to no teeth grinding, with any grinding activity lasting less than 1 second. The Mild Warning recordings contained occasional short grinding episodes lasting approximately 2–5 seconds. The Severe Warning recordings were characterized by frequent grinding episodes, with grinding activity present for at least 80% of the recording duration. The classification pipeline used to assign each recording to a warning level follows below.

- **Signal cleanup:** The raw microphone signal is adjusted so that unwanted background offsets and slow changes are removed.
- **Envelope extraction:** The overall strength of the microphone signal is measured every 50 milliseconds. This helps identify when the sound becomes stronger, such as during possible grinding activity.
- **Thresholding:** A baseline is decided from the quieter parts of the recording. When the signal strength rises above a set threshold, that section is marked as active and considered a possible grinding event.
- **Gap bridging:** Very short gaps between active sections are filled in so that one continuous grinding event is not incorrectly divided into several smaller events.
- **Burst validation:** Each possible grinding event is checked to make sure it is active for a sufficient portion of its duration and has frequency characteristics that are consistent with grinding rather than speech or other sounds.
- **Tier decision:** The validated grinding events are compared with the duration and frequency requirements for each warning level. A warning is only given when enough evidence is present, which helps prevent a single short or unrelated sound from causing a false warning.

TABLE II
Parameters Used by the Filtering/Classification Algorithm

Parameter	Value	Purpose
Grinding frequency band	20–80 Hz	Isolates the frequency range identified from the silence condition grinding analysis as most associated with grinding.
Minimum burst duration (Mild floor)	0.25 s	Filters out short noise blips that are too short to represent a real grinding or clenching event.
Minimum burst duration (Severe floor)	1.5 s	Minimum cutoff for a sustained grinding episode
Burst threshold multiplier	2× baseline	Sets how much louder than the quiet baseline a sample must be to count as “active”.
Baseline percentile	Quietest 8% of envelope	Defines the “quiet” reference level
Gap-bridging window	0.1 s	Combines brief dips inside one continuous grinding stretch so it is not cut up into many small bursts.
Minimum duty cycle	60%	Rejects bridged stretches that are only sparsely active within their own span

Parameter	Value	Purpose
Minimum spectral flatness	0.4	Rejects tonal sounds like speech (flatness near 0) and keeps broadband, noise-like sounds like grinding (flatness near 1).
Minimum corroborating bursts	2	Requires at least 2 qualifying bursts before a Mild or Severe warning fires, so a single stray event cannot trigger a false alarm.

D. Expected Outcome / Hypothesis

The target performance benchmark is $\geq 85\%$ sensitivity and $\geq 85\%$ specificity relative to PSG, which is defined as:

- **Specificity** is avoiding false alarms.
Ex. of 100 non-bruxism instances, ≤ 15 incorrectly flagged
- **Sensitivity** is correctly detecting true events.
Ex. of 50 actual bruxism events, at least 43 correctly detected

This study did not use PSG-labeled ground truth, so formal sensitivity/specificity against PSG could not be found in this round (see Limitations, Section V). Instead, the classification algorithm's tier decisions were validated against the known, self-labeled ground truth of each recorded trial, or trials that are purposefully done to get a specific warning label.

IV. RESULTS

Findings are presented in three groups:

- (1) the microphone placement / occlusion effect trial
- (2) frequency-domain comparison of grinding vs. ambient noise
- (3) output of the filtering/classification algorithm

Fig. 1: Occlusion Effect Trial — Near-Jaw vs. In-Ear Placement

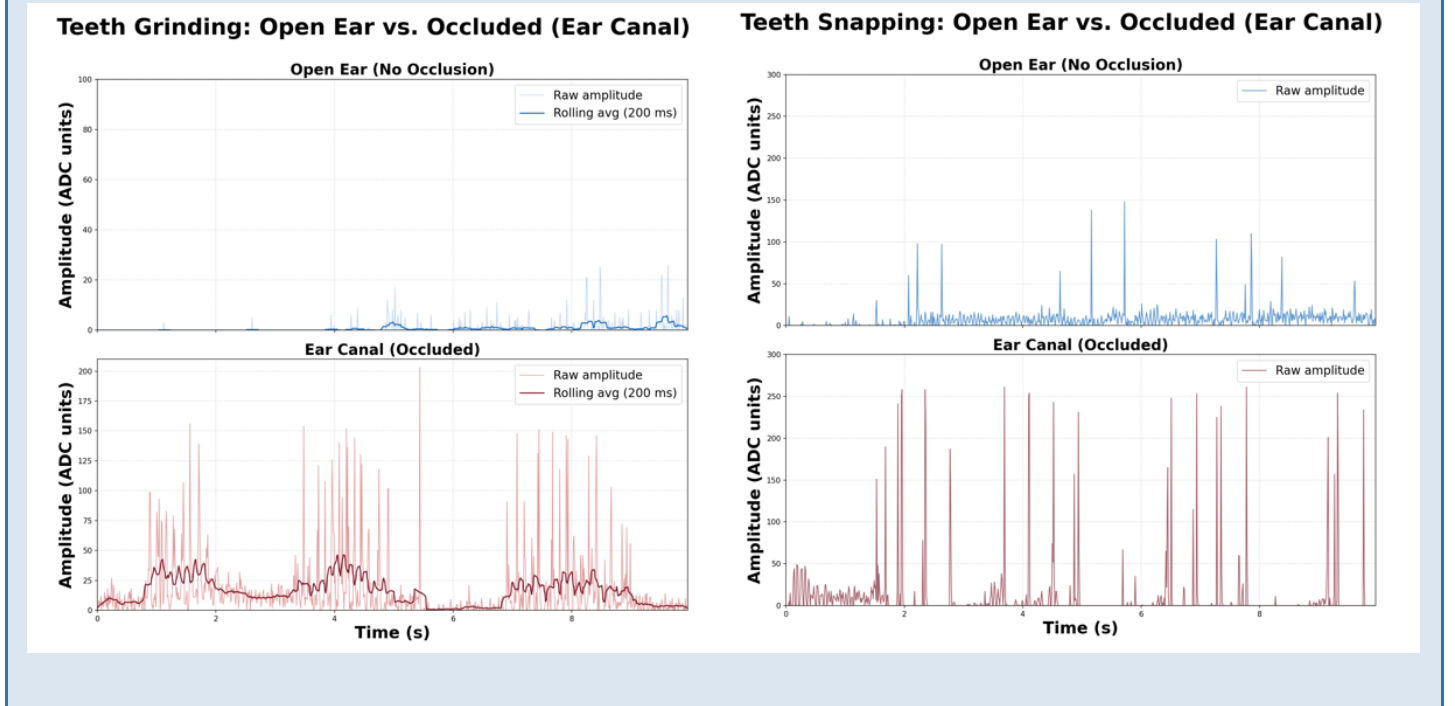


Fig. 1. Comparison of grinding signal amplitudes under each microphone placement

Fig. 2: Baseline Frequency Content — Silence vs. Fan Noise (No Grinding, In-Ear)

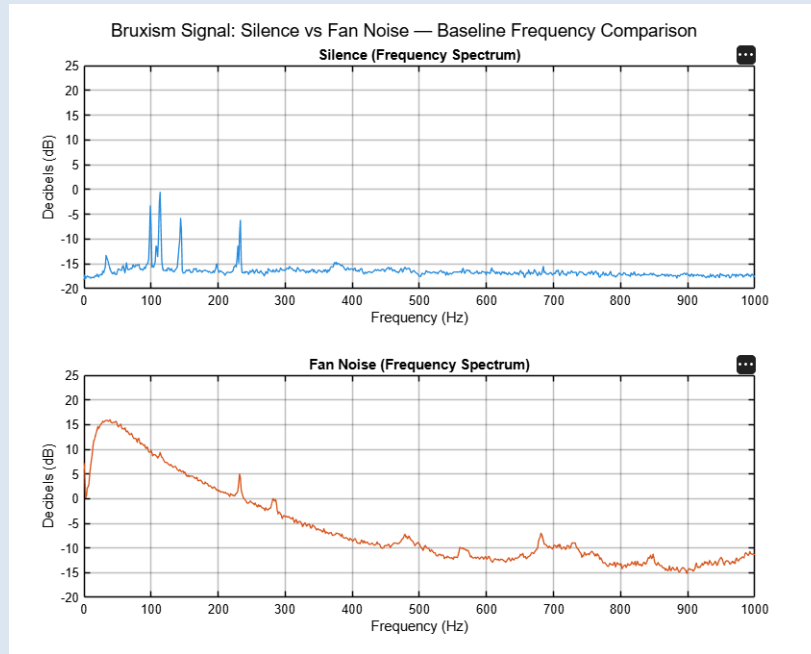


Fig. 2. Frequency domain comparison of 5 minutes of silence and 5 minutes of fan noise, with the microphone in-ear and no grinding. Peaks present here (approximately 110, 140, and 230 Hz, with an additional 285 Hz peak under fan noise) are not related to grinding and are most likely powerline/electrical interference and its harmonics and can be dismissed as non-grinding artifacts.

Fig. 3: Teeth Grinding Frequency Content — Silence vs. Fan Noise (In-Ear)

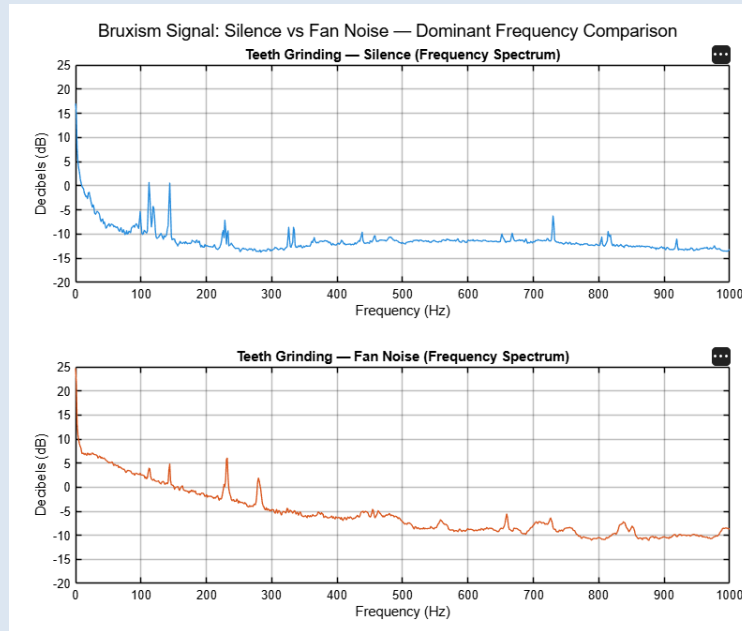


Fig. 3. Frequency domain comparison of 30 minutes of teeth grinding recorded in silence and 30 minutes recorded under fan noise. Peaks shared with the no grinding baseline (Fig. 2) are treated as non-grinding artifacts. The 20–80 Hz band is distinct between grinding and non-grinding activity and was used as the target band for the filtering algorithm (Section III).

Fig. 4: Filtering Algorithm Output — No Warning Trial

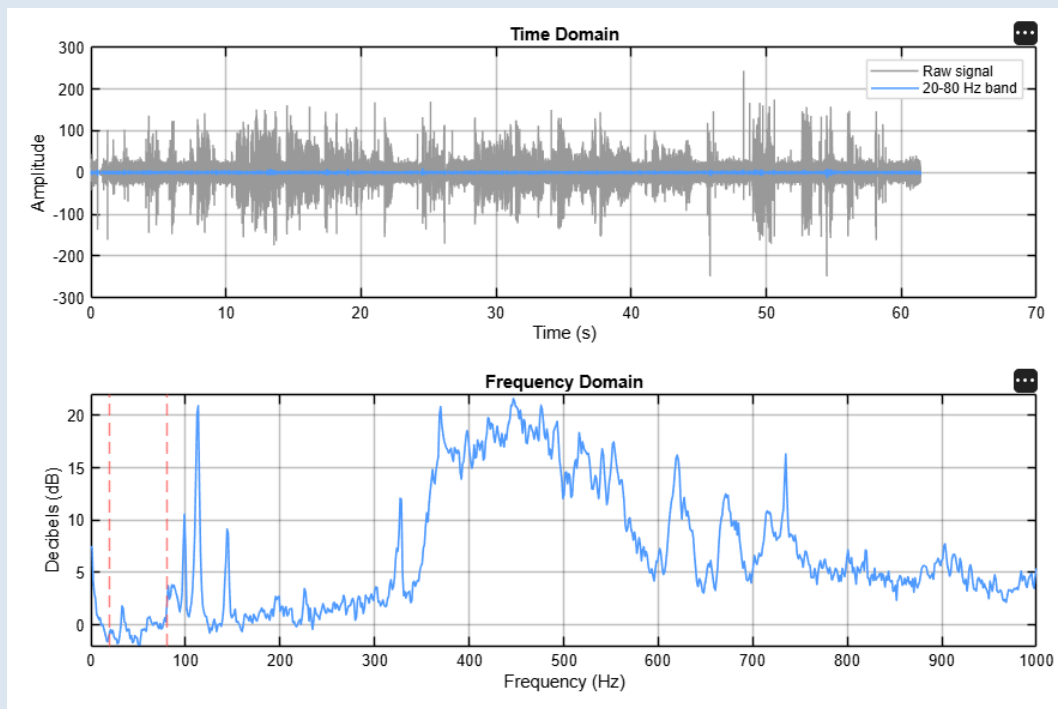


Fig. 4. Example output of the filtering algorithm on a No Warning trial.

Fig. 5: Filtering Algorithm Output — Mild Warning Trial

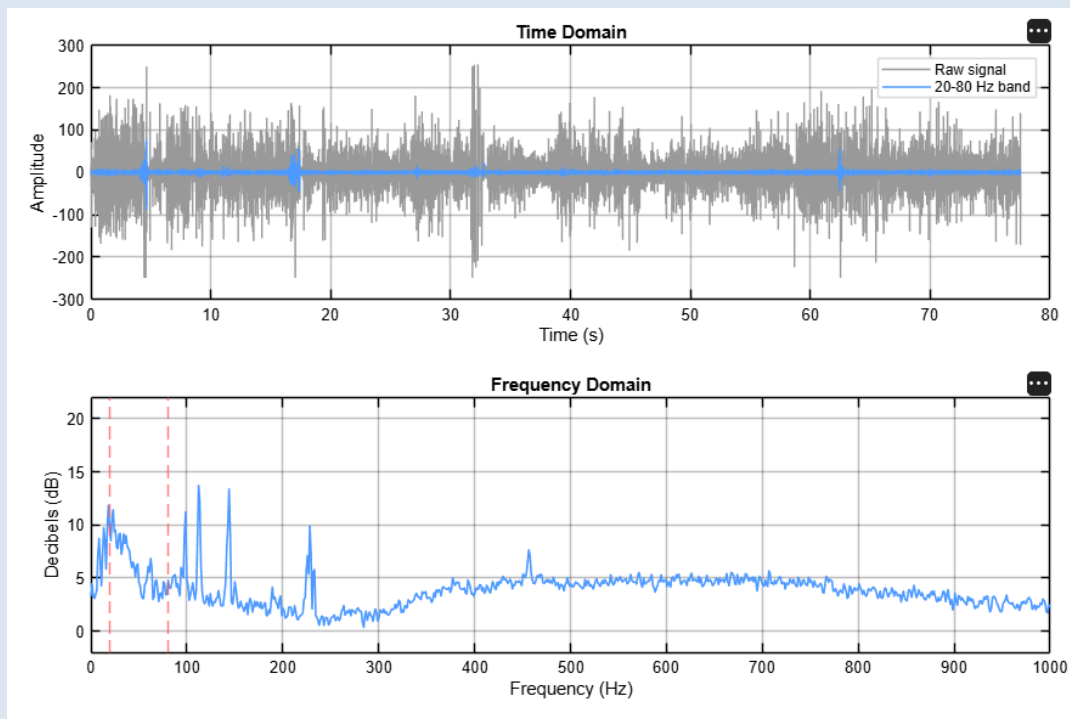


Fig. 5. Example output of the filtering algorithm on a Mild Warning trial.

Fig. 6: Filtering Algorithm Output — Severe Warning Trial

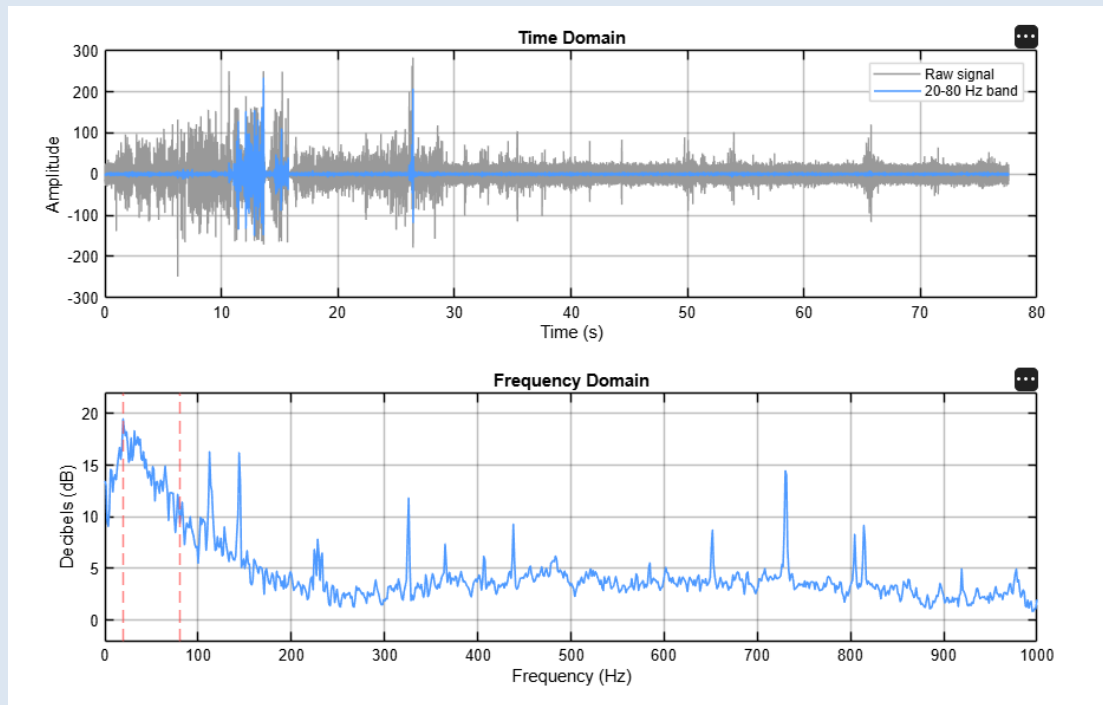


Fig. 6. Example output of the filtering algorithm on a Severe Warning trial

Classification Statistics

The filtering algorithm correctly classified every trial test. Table III summarizes the average statistics the algorithm computed for each tier.

TABLE III
Averaged Classification Statistics by Warning Tier

Statistic	No Warning (Avg, Trials 1–3)	Mild (Avg, Trials 1–3)	Severe	What It Measures & Why It Matters
Recording Duration (s)	72.3 s	77.5 s	77.5 s	Average length of the analyzed recordings
Baseline Envelope Level	7.46	7.37	7.62	Quietest 8% of the recording; noise floor as reference point.
Detection Threshold (Amplitude)	14.93	14.73	15.25	Envelope level a sample must exceed to be flagged as an active, possible-grinding sample.
Total Active Time in Grinding Band	32.5 s	12.0 s	29.8 s	Total time the envelope stayed above threshold; raw measure of loud activity before burst-validity filtering.
Active Time Percentage	36.1%	15.3%	38.3%	Active time from the whole recording, making trials of different lengths directly comparable.
Number of Bursts Clearing Mild Length Floor (≥ 0.25 s)	0.3	7.0	10.7	Count of active stretches long enough to be a real event rather than noise. Feeds into the Mild/Severe decision.
Number of Bursts Clearing Severe Length Floor (≥ 1.5 s)	0	0.3	2.7	Count of sustained bursts meeting the duration cutoff for a bruxism episode; feeds into the Severe decision.
Longest Continuous Grinding Activity	0.16 s	1.39 s	4.04 s	Duration of the single longest qualifying burst — captures the most intense episode, which the total active-time figure alone can hide.

One notable pattern in Table III is that the No Warning trials show more raw active time (32.5 s) than the Mild trials (12.0 s). This is expected, because that statistic counts all envelope-threshold crossings before burst-validity filtering is applied, so it can be inflated by non-grinding activity like speech. The burst-count and duration statistics computed after duty-cycle and spectral-flatness filtering increase consistently from No Warning to Mild to Severe, which is why the algorithm's tier decision is based on those validated burst statistics rather than on raw active time alone.

More generally, the absolute values the algorithm reports for detected grinding time do not precisely match the actual physical grinding time in each trial. However, the validated burst statistics still follow the correct relative trend across tiers (increasing from No Warning to Mild to Severe), which is what the tier-classification logic relies on, and is consistent with the 3/3 classification accuracy reported above.

V. DISCUSSION

A. Interpretation of Results

- Teeth grinding produced clear, distinguishable acoustic signatures, confirming that jaw-generated sounds can be reliably captured near or within the ear.
- Occluding the ear canal with an in-ear microphone increased grinding-signal amplitude by roughly 10× relative to a near-jaw, non-occluded placement, directly confirming the occlusion-effect prediction from Section II and determining the switch to in-ear placement for all subsequent testing.
- The 20–80 Hz band, isolated after excluding peaks shared with the no-grinding baseline, was identifiable under both silence and fan-noise conditions.
- Although the the algorithm's active-time and burst-duration values did not align to the ground truth, the relative trend follows across the tiers and shows feasibility for a more accurate model.

Together, these results indicate that ear-canal occlusion improves acoustic sensitivity to grinding activity specifically, making occlusion a useful design strategy for wearable, ear-level bruxism sensing.

B. Comparison to Previous Studies

Unlike the EMG-based devices reviewed in Section II, this approach avoids the electrode-placement reliability problem that limits EMG devices during overnight use. Because the system uses low-cost, off-the-shelf components it also shows stronger potential for accessibility than PSG or the reviewed EMG-based alternatives, which require either a clinical setting or specialized hardware. This study's empirical result provides a quantified confirmation in the specific context of teeth-grinding detection using acoustic signals.

C. Implications & Limitations

This approach shows strong potential for accessible, low-cost, early detection of bruxism outside of clinical sleep lab environments, and could plausibly extend toward broader sleep-event screening using the same acoustic sensing pipeline.

A major limitation of this study was the sample size as a single subject ($N = 1$) throughout. All trials were classified correctly for this subject, but this is not enough evidence that the same

frequency band, thresholds, or accuracy will generalize to other people. Another major limitation is tied to the acoustic signal approach itself, as the current system does not detect non-acoustic clenching, which is a clinically relevant subtype of bruxism. Due to the controlled settings of this study, filtering out movement artifacts from the target grinding signal remains a challenge not yet fully solved. Furthermore, no PSG-labeled ground truth was used, so sensitivity/specificity against the clinical reference standard has not yet been measured and will serve as a critical point of concern for the future of this study. Other practical constraints, such as user comfort, battery life, and sufficient clinical testing are prominent research problems before this could move toward a deployable product.

VI. CONCLUSION

A. Summary

This study found that teeth grinding produces acoustic signals that are reliably detectable within the ear canal, with the in-ear (occluded) microphone placement improving grinding signal capture by roughly 10× relative to a near-jaw placement. After identifying a 20–80 Hz grinding-associated frequency band and excluding peaks shared with non-grinding baseline recordings, a MATLAB-based filtering algorithm using envelope thresholding, gap-bridging, duty-cycle, and spectral-flatness criteria correctly classified all tested trials into the correct warning tier for this subject. Formal sensitivity and specificity against a PSG reference standard were not evaluated in this round and remain a target for future work.

B. Contributions

This work demonstrates a low-cost, non-invasive, audio-only alternative to existing largely EMG-based bruxism screening devices, addressing a gap identified in Section II, since none of the devices reviewed use acoustic sensing as their primary detection method. It also provides an early, quantified result for how ear-canal occlusion affects signal capture, which can inform future earable sensor design beyond bruxism specifically, along with a working, fully documented filtering algorithm and parameter set that can be extended or re-tuned for future subjects.

C. Future Research

- Validate the filtering algorithm and frequency band on additional subjects to assess generalizability beyond N = 1.
- Measure true sensitivity and specificity against the $\geq 85\%$ / $\geq 85\%$ benchmark (Section III).
- Use pattern recognition or machine learning algorithms to efficiently flag irregularities given a more diverse dataset.
- Refine the prototype for improved comfort, size, and consistency to support reliable overnight use.
- Run tests in natural (home) conditions, with a larger sample, to assess feasibility beyond controlled lab testing.

VII. REFERENCES

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